

VED UMRAJKAR

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Education

Indian Institute of Technology, Roorkee

2021 – 2026

Bachelors and Masters, Mathematics and Computing

CGPA: 8.605/10.0

Coursework: Statistical Machine Learning, Natural Language Processing, Stochastic Calculus, Design and Analysis of Algorithms, Mathematical Cryptography

Experience

PavoAI | *Research Intern*

Feb 2026 – Present

Remote

- Co-authored “**Agentic Evidence Selection for Decision-Making under Cognitive Constraints**” (KDD 2026 Research Track) [Paper], formalising task-agnostic evidence selection as a **constrained submodular optimisation** problem.
- Designed a framework combining **semantic anchor-based importance scoring** and diversity-aware selection, achieving **95.4%** of theoretical maximum semantic coverage without any task supervision.
- Showed that importance-based pre-filtering retains **98% semantic coverage using only the top 20%** of evidence, generalising the pipeline to corpus-based retrieval architectures.

JPMorgan Chase | *Intern*

May 2025 – Jul 2025

Bengaluru

- Implemented the core logic for a macroeconomic-indicator-driven **investment management fees forecasting model** and deployed it to production within the Data Science CoE.
- Built granular **macroeconomic forecasting prototypes** with headcount sensitivity analysis to account for COVID-driven disparities and stress scenarios.

Hashstack Finance | *Data Scientist*

Aug 2023 – Feb 2024

Remote

- Built a real-time **risk management and analytics framework** with Starknet, PostgreSQL, and Streamlit to model debt health and solvency metrics.
- Integrated mean-variance **portfolio optimization** into the backend to flag impermanent-loss risk, delivering a **15% lift in outlier detection accuracy**.
- Designed a unified alert system and **growth analytics dashboard** integrating Mixpanel, Google Analytics, and on-chain Starknet data via Streamlit and Python.

Research & Projects

Tree Ring Watermark Detection in Rectified Flow Models [Code] [Paper]

Jan 2025 – Apr 2025

- Benchmarked Tree Ring watermark detectability and false-positive rates on DDIM (Stable Diffusion 2.1) vs **rectified flow transformers** (FLUX.1-dev) using standardized perturbation budgets.
- Diagnosed inversion instability in rectified flows (FlowEuler, OT-CFM) and proposed robust watermark recovery settings for **text-to-image safety pipelines**.
- Presented empirical limits, dataset curation, and statistical separability results at **ICLR 2025**, Singapore Expo.

DAC-LoRA: Dynamic Adversarial Curriculum for VLMs [Code] [Paper]

2025

- Introduced **DAC-LoRA**: LoRA adapters trained with progressively stronger PGD attacks to harden CLIP-style VLMs, presented at the *Workshop on Safe Multimodal AI Systems, ICCV 2025*.
- Balanced **robust accuracy** and adaptation cost via small-rank adapters and curriculum pacing, outperforming vanilla LoRA and single-stage adversarial fine-tuning on few-shot benchmarks.

Multi-Modal (Video-Text) Person Re-Identification [Code]

Dec 2024 – Present

- Building a video-text re-ID stack with tracklet encoders and a transformer text encoder, aligned via **contrastive and triplet losses** for open-vocabulary retrieval.
- Running **ablation studies** on 5 benchmarks (MEVID, LSVID, ILIDS, Duke, G2A); reporting mAP/CMC gains from temporal pooling, query expansion, and hard-negative mining (ongoing).

Voice Cloning with AutoRegressive Acoustic Models [Code]

Mar 2024 – Apr 2024

- Developed automated voice clones converting text into discretized acoustic tokens using **Tortoise-TTS**, tuning softmax temperature and mel spectrogram synthesis for high-fidelity output.
- Built a **Streamlit** app for real-time voice clone generation, allowing users to upload **.wav** files to create custom voices.

Multi-Risk Portfolio Optimization [\[Code\]](#)

Feb 2024 – Apr 2024

- Constructed S&P 500 portfolios with **Riskfolio**, maximizing Sharpe ratio while enforcing turnover and sector exposure constraints to keep allocations realistic.
- Stress-tested allocations using standard deviation, **CVaR**, **MDD**, and **MAD** to quantify tail risk and drawdown resilience.
- Ran **Monte Carlo bootstraps** and log-normal simulations for forward-looking return bands and downside probability estimates.

ForecastPro [\[Code\]](#)

Mar 2023 – Apr 2023

- Built a hybrid forecaster combining **LSTNet** with classical **ARIMA/SARIMAX** baselines to cover seasonal, trend, and sudden-shift patterns.
- Engineered lagged covariates, holiday flags, and rolling statistics; tuned ConvLSTM windows to stabilize long-horizon predictions.
- Evaluated on AirQuality UCI and Yahoo Finance with **MAPE**/RMSE, showing consistent lift over pure statistical models.

Awards / Scholarships

- Silver Medal at Inter-IIT Tech Meet 13.0 at IIT Bombay – Built **AI-powered Product** for Dream11.
- Secured **16th Rank** in **Amazon ML Challenge 2024** (10,000+ candidates).
- Qualified for India Terminal Coding competition hosted by Correlation One in partnership with Citadel Securities.
- All India Rank **342** in KVPY Fellowship Examination.
- State Topper from Maharashtra in Astronomy Olympiad (IOQA-2020).
- Rated **Expert on Kaggle** (top 1%, Global Rank 1890, handle: vedumrajkar).
- Awarded the INSPIRE Scholarship by the Government of India for undergraduates in research.
- All India Rank **1336** in JEE Advanced (**Top 0.2%** of all candidates).

Volunteering

Data Science Group, IIT Roorkee | *Joint Secretary*

Dec 2023 – May 2025

- Mentored students pursuing careers in **machine learning**, fostering a campus community around applied AI.
- Coordinated campus projects and events, hosting **data science workshops** and managing member recruitment.
- Hosted a two-week **supervised learning** hackathon on the UCI Gesture Phase Segmentation dataset.

Technical Skills

- **Areas of Interest:** Generative AI (GANs, Diffusion Models), Large Language Models (LLMs), Machine Learning, Computer Vision, Multimodal Learning
- **Languages:** Python, C++, SQL
- **Frameworks/Libraries:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, NLTK, Hugging Face, Streamlit